Jacob McLaughlin

mclaughlinjacob2005@gmail.com | 925-758-9193 | linkedin.com/in/jacob-mcl | github.com/BacoJaco | jacobmcl.dev

Education

University of California, Riverside | BS in Computer Science – GPA: 3.90/4.00 Expected June 2028

- **Awards and Honors:** Tau Beta Pi Engineering Honors Society, Dean's Honor List
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Software Engineering & Construction, Discrete Structures

Projects

LiDAR Autonomous Delivery Robot | ACM @ UCR: Forge – Riverside, CA Apr 2026 – Present

- Developing the autonomy stack for a differential-drive robot designed for food delivery with 4 team members
- Implementing A* path planning and GPS-tracking to compute optimal, collision-free trajectories in real time
- Designing and integrating 2D LiDAR-based mapping and localization with ROS-based environment perception
- Integrating a YOLOv8 computer vision model to detect and dynamically avoid humans

University Web Applications | ACM @ UCR: Spark – Riverside, CA Jan 2026 – Present

- Developing a Next.js web application within a team of 8 for the UCR Chapter of University Blood Initiative, leveraging SSR (Server-Side Rendering) to reduce initial load times by up to 70% and improve SEO visibility
- Implementing a responsive, mobile-first UI using Tailwind CSS, improving usability across desktop and mobile devices for 50+ student members
- Developed the Board Members, Events, and Get Involved pages for the UCR Archery Club web app with fully mobile-responsive layouts, serving 30+ members

Object Detection Surveillance System | ACM @ UCR: Forge – Riverside, CA Jan 2026 – Apr 2026

- Built a real-time multi-class detection system using YOLOv8, achieving over 90% accuracy across people, vehicles, and common objects in dynamic environments
- Designed an MQTT messaging pipeline handling 30+ messages/sec for real-time perception output
- Integrated the computer vision model with an ESP32 camera module, enabling live remote monitoring

Terminal-Based Crossy Road | github.com/BacoJaco/terminalcrossy Sep 2025 – Dec 2025

- Designed a replica of Crossy Road with 3 team members, leveraging SOLID principles in C++
- Implemented agile methodologies using Kanban boards, UML diagrams and standup meetings to optimize development speed and team synchronization
- Deployed CI/CD pipeline using Google Test to ensure 99% build stability during active development

Leadership Experience

Lead Computer Vision Engineer | ACM @ UCR: Forge – Riverside, CA Apr 2026 – Present

- Leading 10 members across 5 sub-teams to deliver an AI-powered surveillance system, coordinating Python and Next.js development to launch a functional prototype 2 weeks ahead of schedule
- Mentoring team members on YOLOv8, Python multithreading, and MQTT to accelerate technical onboarding
- Overseeing the development of a Next.js dashboard to display live camera feeds and YOLOv8 detections

Lead Game Developer | GameSpawn – Riverside, CA Sep 2025 – Jan 2026

- Led a team of 3 developers to design and build a multi-level Unity game, using Agile sprints and Git for version control, and set technical direction that enabled on-time delivery
- Developed 7 interactive levels using C#, implementing core gameplay mechanics and navigation systems
- Designed the game to support both virtual reality (Oculus) and PC platforms, ensuring cross-platform compatibility and expanding the potential player base

Skills

Languages: C++, Python, C#, HTML, TypeScript, JavaScript, Tailwind CSS

Tools & Frameworks: Git, React, Next.js, Node.js, Unity, ROS, Google Test